

$$8 = 2^3$$

$$\log_2 8 = 3$$

## Week 6 - Part 2 - Osprey1st College Algebra - Summer 2024

### Section covered: 9.3, 9.4, and 9.5

The horizontal line test shows that exponential functions are one-to-one and thus have inverse functions. The equation defining the inverse of a function is found by interchanging  $x$  and  $y$  in the equation that defines the function. Starting with  $y = a^x$  and interchanging  $x$  and  $y$  yields  $x = a^y$ .

#### Logarithm

For all real numbers  $y$  and all positive numbers  $a$  and  $x$ , where  $a \neq 1$ ,

$$y = \log_a x \text{ is equivalent to } x = a^y.$$

The expression  $\log_a x$  represents the exponent to which the base  $a$  must be raised in order to obtain  $x$ .

Problem 12. \* Evaluate the given logarithms.

(a)  $\log_5 25 = x$   $5^x = 25$   $x = 2$

(b)  $\log_2 8 = x$   $2^x = 8$   $x = 3$

(c)  $\log_3 1 = x$   $3^x = 1$   $x = 0$

(d)  $\log_4 \frac{1}{4} = x$

$4^x = \frac{1}{4}$   $x = -1$   $4^{-1} = \frac{1}{4}$

(e)  $\log_3 \frac{1}{9} = x$

$3^x = \frac{1}{9}$   $x = -2$   $3^{-2} = \frac{1}{3^2} = \frac{1}{9}$

Problem 13. \* Rewrite each equation as requested.

| Logarithmic Equation        | Exponential Equation     |
|-----------------------------|--------------------------|
| $\log_2 32 = 5$             | $2^5 = 32$               |
| $\log_6 \frac{1}{36} = -2$  | $6^{-2} = \frac{1}{36}$  |
| $\log_5 \frac{1}{125} = -3$ | $5^{-3} = \frac{1}{125}$ |
| $\log_{12} 12 = 1$          | $12^1 = 12$              |
| $\log_6 1 = 0$              | $6^0 = 1$                |

$10^x = 100$   $x = 2$

$\log_{10} 100 = x$

#### Common Logarithms

Two of the most most important bases for logarithms are 10 and  $e$ . Base 10 logarithms are called common logarithms. The common logarithm of  $x$  is written  $\log x$  where the base is understood to be 10.

### Common Logarithm

For all positive numbers  $x$ ,

$$\log x = \log_{10} x.$$

### Natural Logarithm

The number  $e$  is defined as the value that  $\left(1 + \frac{1}{n}\right)$  approaches as  $n$  gets larger and larger, that is

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \approx 2.718281828459$$

The irrational number  $e$ , approximately 2.72, is called the **natural base**.

Logarithms with base  $e$  are natural logarithms because they occur in the life sciences and economics in natural situations that involve growth and decay. The base  $e$  logarithm of  $x$  is written **ln  $x$**  (read “el-en  $x$ ”). The expression  $\ln x$  represents the exponent to which  $e$  must be raised in order to obtain  $x$ .

### Natural Logarithm

For all positive numbers  $x$ ,

$$\ln x = \log_e x.$$

Problem 14. Evaluate the given logarithms.

(a)  $\log 1000 = x$   $10^x = 1000$   $x = 3$  (c)  $\ln 1 = x$   $e^x = 1$   $x = 0$   
(b)  $\log \frac{1}{100} = x$   $10^x = \frac{1}{100}$   $x = -2$  (d)  $\ln e = x$   $e^x = e$   $x = 1$

Problem 15. \* Rewrite each equation as requested.

(a) Rewrite as an exponential equation.

$$e^7 = x \quad \ln x = 7$$

(c) Rewrite as an exponential equation.

$$e^y = 4 \quad \ln 4 = y$$

(b) Rewrite as a logarithmic equation.

$$e^y = 2$$

$$\ln 2 = y$$

(d) Rewrite as a logarithmic equation.

$$e^6 = x$$

$$\ln x = 6$$

# Logarithmic Functions

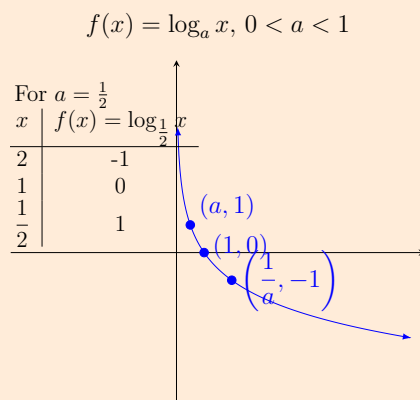
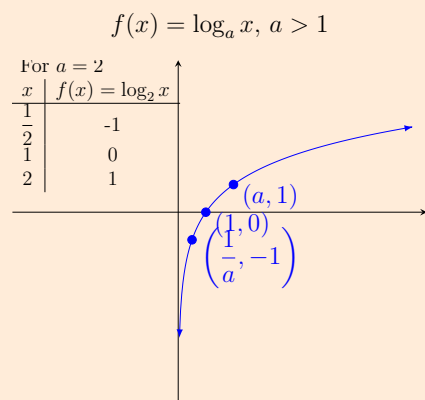
## Logarithmic Function

If  $a > 0$ ,  $a \neq 1$ , and  $x > 0$ , then the **logarithmic function with base  $a$**  is

$$f(x) = \log_a x.$$

## Characteristics of the Graph of $f(x) = \log_a x$

1. The point  $(\frac{1}{a}, -1)$ ,  $(1, 0)$  and  $(a, 1)$  are on the graph.
2. If  $a > 1$ , then  $f$  is an increasing function. If  $0 < a < 1$ , then  $f$  is a decreasing function.
3. The  $y$ -axis is a vertical asymptote.
4. The domain is  $(0, \infty)$  and the range is  $(-\infty, \infty)$ .



## Properties of Logarithms

For  $x > 0$ ,  $y > 0$ ,  $a > 0$ ,  $a \neq 1$ , and any real number  $r$ , the following properties hold.

**Product Rule**  $\log_a xy = \log_a x + \log_a y$

**Quotient Rule**  $\log_a \frac{x}{y} = \log_a x - \log_a y$

**Power Rule**  $\log_a x^r = r \log_a x$

**Logarithm of 1**  $\log_a 1 = 0$

**Base  $a$  Logarithm of  $a$**   $\log_a a = 1$

**Note:** There is no property of logarithms to rewrite a logarithm of a sum or difference.

Problem 16. \* Fill in the missing values to make the equations true.

(a)  $\log_8 5 - \log_8 11 = \log_8 \square$

(b)  $\log_9 \square + \log_9 12 = \log_9 60$

(c)  $\log_6 25 = \square \log_6 5$

Problem 17. \* Rewrite each expression. Assume all variables represent positive real numbers, with  $a \neq 1$  and  $b \neq 1$ .

(a)  $\log_7(8 \cdot 6)$

(b)  $\log_6 \frac{5}{9}$

(c)  $\log_a \sqrt[3]{m^2}$

(d)  $\log \left( \frac{\sqrt[3]{x^7 z}}{y^2} \right)$

(e)  $\log_a \frac{x^5}{z^4 \sqrt[5]{y^3}}$

(f)  $\log \sqrt[3]{\frac{xy^4}{z^2}}$

(g)  $\log_4(x^2 + y^2)$

Problem 18. \* Write each expression as a single logarithm with coefficient 1. Assume all variables represent positive real numbers, with  $a \neq 1$  and  $b \neq 1$ .

(a)  $\log_4 x - \log_4 y + \log_4 z$

(b)  $\frac{1}{3} \log_a x + \frac{2}{3} \log_a y - \log_a xy$

(c)  $6 \log_m (3z + 1) + \frac{1}{4} \log_m (z + 7)$

(d)  $4(\log_c y - 4 \log_c w) + 3 \log_c x$

(e)  $3 \log_m x + 2(\log_m w - 2 \log_m y)$

Problem 19. \* Use the properties of logarithms to evaluate each of the following expressions.

(a)  $\log_2 36 - 2 \log_2 3 =$

(b)  $\ln e^{-8} + \ln e^3 =$

(c)  $2 \ln_5 3 - \log_5 45 =$

## Logarithms with Other Bases

We can use a calculator to find the values of either natural logarithms (base e) or common logarithms (base 10). However, sometimes we must use logarithms with other bases. The change-of-base theorem can be used to convert logarithms from one base to another.

### Change-of-Base Theorem

For any positive real number  $x, a$  and  $b$ , where  $a \neq 1$  and  $b \neq 1$ , the following holds.

$$\log_a x = \frac{\log_b x}{\log_b a}.$$

In particular,

$$\log_a x = \frac{\log x}{\log a} \quad \text{or} \quad \log_a x = \frac{\ln x}{\ln a}.$$

Problem 20. Use the change-of-base theorem to write each logarithm (i) in terms of common logarithms, (ii) in terms of natural logarithms. Approximate the answer to four decimal places for each logarithm.

(a)  $\log_4 20$

(b)  $\log_2 0.7$

### Solving an Exponential or Logarithmic Equation

To solve an exponential or logarithmic equation, change the given equation into one of the following forms, where  $a$  and  $b$  are real numbers,  $a > 0$  and  $a \neq 1$ , and follow the guidelines.

1.  $a^{f(x)} = a^{g(x)}$

The given equation is equivalent to the equation  $f(x) = g(x)$ . Solve algebraically.

2.  $a^{f(x)} = b$

Solve by taking natural (or common) logarithms on each side  $\ln a^{f(x)} = \ln b$  and then bring  $f(x)$  in front of the logarithm,  $f(x) \ln a = \ln b$ .

3.  $\log_a f(x) = b$

Solve by changing to exponential form  $a^b = f(x)$ . Check that each proposed solution is in the domain.

4.  $\log_a f(x) = \log_a g(x)$

The given equation is equivalent to the equation  $f(x) = g(x)$ . Solve algebraically. Check that each proposed solution is in the domain.

## Exponential equations

Problem 1. \* Solve each exponential equation

(a)  $64 = 32^{-x+2}$

(b)  $10^{-8x} = 100^{2-3x}$

(c)  $36^{3x-5} = 6^{8x}$

Problem 2. \* Solve each equation. Give exact values using either base-10 or base- $e$  logarithms. Then round your answer to the nearest hundredth. Do not round any intermediate computations.

(a)  $5^{-6x} = 15$

(b)  $2^{x+8} = 3$

(c)  $2^{-x+10} = 5^{-6x}$

(d)  $13^{-x+10} = 4^{2x}$